

Q1. • Squared Error loss per sample is:

$$\text{Squared Error} = (y_{\text{actual}} - y_{\text{predicted}})^2$$

y_{actual} = ground truth
 $y_{\text{predicted}}$ is model's prediction.

1. $(2-3)^2 = 1^2 = 1$

2. $(4-5)^2 = 1^2 = 1$

3. $(6-5)^2 = 1^2 = 1$

• MSE, Mean Squared Error = $\frac{1}{N} \sum_{i=1}^N (y_{\text{actual},i} - y_{\text{predicted},i})^2$

$N=3$

Calculate:

$$\text{MSE} = \frac{1+1+1}{3} = \frac{3}{3} = 1$$

Ans. MSE = 1

Q2.

Step-1: Expression for Predictor Function, $f(x)$

Step-2: Identify Loss for Each Example

$$\text{Absolute Error} = |y_{\text{actual}} - y_{\text{predicted}}|$$

Actual:

Predicted:

1. $|10-12| = |1-2| = 2$

2. $|20-18| = |2| = 2$

3. $|30-33| = |1-3| = 3$

Step-3 : Expression for Cost Function (MAE)

$$\text{Mean Absolute Error, (MAE)} = \frac{1}{N} \sum_{i=1}^N |y_{\text{actual},i} - y_{\text{predicted},i}|$$

Where $N=3$.

Calculate:

$$\text{MAE} = \frac{2+2+3}{3} = \frac{7}{3} \approx \underline{2.333}$$

Step-4 : Gradient of Cost Function

Calculation is direct since no weights are updated in this evaluation.

Final Answer : $\text{MAE} = \underline{2.333}$

◦ Each Absolute Error :

◦ Average over all errors : $\frac{7}{3} = 2.333$

Q3. Step-1 : Expression for Predictor Function

◦ Output is a predicted probability 'p' from the model, using sigmoid activation (range).

◦ Here $p=0.2$.

Step-2 : Identify the Loss to use for Each Training Example :

◦ The binary cross-entropy (log-loss) is:

$$\text{Loss} = -[y \cdot \log(p) + (1-y) \cdot \log(1-p)]$$

Where:

◦ y = true label (0 or 1)

◦ p = predicted probability for class 1.

Step-3 : Cost Function Expression :

◦ For one sample, the cost is just the above loss.

Step 4: Substitute the Given values:-

- $y = 0, p = 0.2$

- Compute:

$$\text{Loss} = -[0 \cdot \log(0.2) + (1-0) \cdot \log(1-0.2)]$$
$$= -[\log(0.8)]$$

- $\log(0.8) \approx -0.2231$

- $-(-0.2231) = 0.2231$

Step 5: Interpretation:-

- This measures the penalty for predicting probability 0.2 for true label 0 - a lower loss means better alignment with the label.

Final Answer:

- Binary Cross-Entropy Loss ≈ 0.2231 for this sample.

Q4. Step-1: Expression for Predictor Function

- Output is a vector of predicted probabilities for 3 classes: $[0.6, 0.3, 0.1]$ from softmax layer.

- True label is $\text{classI}(\text{Pog})$. The correct probability is $p_1 = 0.3$.

Step 2: Identify the Loss

- For categorical cross-entropy, using one-hot encoding c for the label:

$$\text{Loss} = - \sum_{i=1}^n y_i \log(p_i)$$

Where:

- $y_i = 1$ for true class, 0 otherwise.

- p_i = predicted probability for class i .

Step-3: Cost function Expression

- For this sample, only class 1 is nonzero:
$$\text{Loss} = -(0.6 \log(0.6) + 1 \cdot \log(0.3) + 0 \cdot \log(0.1))$$
$$= -\log(0.3)$$

Step-4: Substitute given values

- $\log(0.3) \approx -1.2040$
- $-(-1.2040) = \underline{1.2040}$

Step-5: Interpretation

This loss penalizes the predicted probability (0.3) for the correct class (Dog) when it should be close to 1.

Final Answer: Categorical Cross-Entropy Loss $\approx \underline{1.2040}$ for Dog Class

Q5. Hinge Loss (True Label = -1, Predicted Score = -0.8)

Step 1: Predictor Function

- Model outputs a raw score, not a probability (typically for SVM).

$$\text{score} = -0.8$$

- True label $y = -1$

Step 2: Loss for each Example

- Hinge Loss:

$$\text{Loss} = \max(0, 1 - y \cdot \text{score})$$

Step 3: Cost Function Expression

- For one sample, use above formula.

Step 4: Substitute values

- $y \cdot \text{score} = (-1) \cdot (-0.8) = 0.8$
- $1 - 0.8 = 0.2$
- $\max(0, 0.2) = 0.2$

Step 5: Interpretation

◦ Hinge Loss penalizes misclassification and lack of margin (here margin is 0.2).

Final Answer for Q5:

Hinge Loss = 0.2 for this sample.